Class 11: AlphaFold

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~250 thousand structures (we saw this in the last lab). The main protein sequence database has over **200 million** sequences! Only 0.125% of known sequences have known structure - this is called the "structure knowledge gap".

```
(250000/200000000)*100
```

```
[1] 0.125
```

- Structures are much harder to determine than sequences
- They are expensive (on average ~$1 million each)
- They take on average 3-5 years to solve!

EBI AlphaFold Database

The EBI has a database of pre-computed AlphaFold(AF) modles called AFDB. This is growing all the time and can be useful to check before running AF ourselves.

Running AlphaFold

We can download and run locally (on our own computers) but we need a GPU. Or we can use "cloud" computing to run this on someone elses computers :-)

We will use ColabFold < https://github.com/sokrypton/ColabFold >

We previously found there was no AFDB entry for our HIV sequence:

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQ
```

Here we will use AlphaFold2_mmseqs2

Custom analysis of resulting models

```
results_dir <- "hivpr_23119/"
```

```
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100

                            101        .         .         .         .         150
[Truncated_Name:1]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivpr_2311  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
```

```
                                 **************************************************
                               101        .         .         .         .         150

                               151        .         .         .         .       198
[Truncated_Name:1]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                                 **************************************************
                               151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

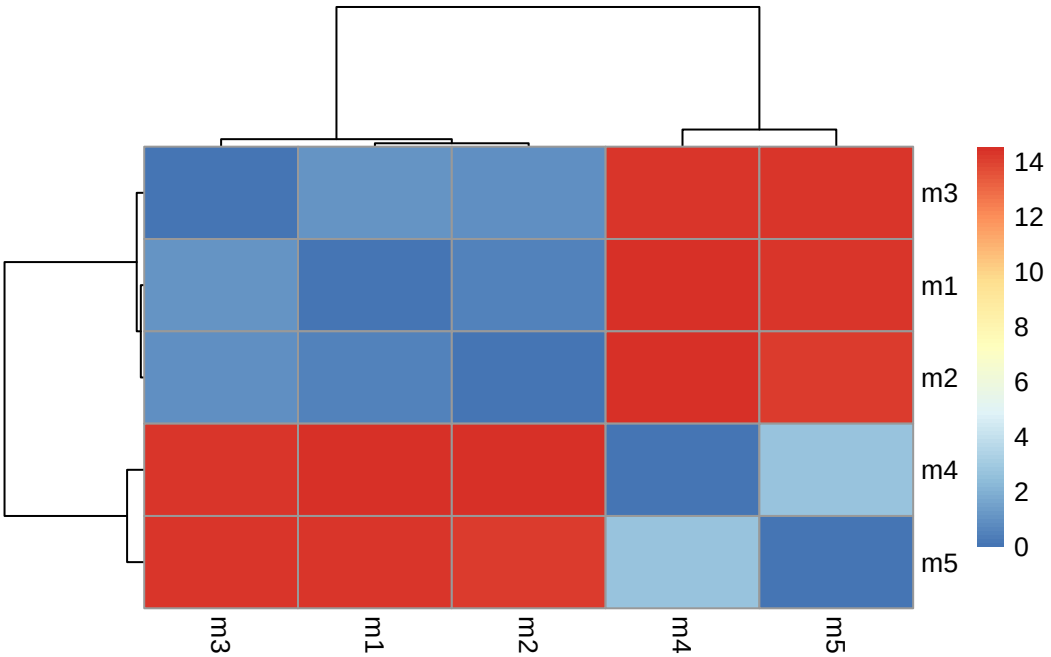
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.539
```

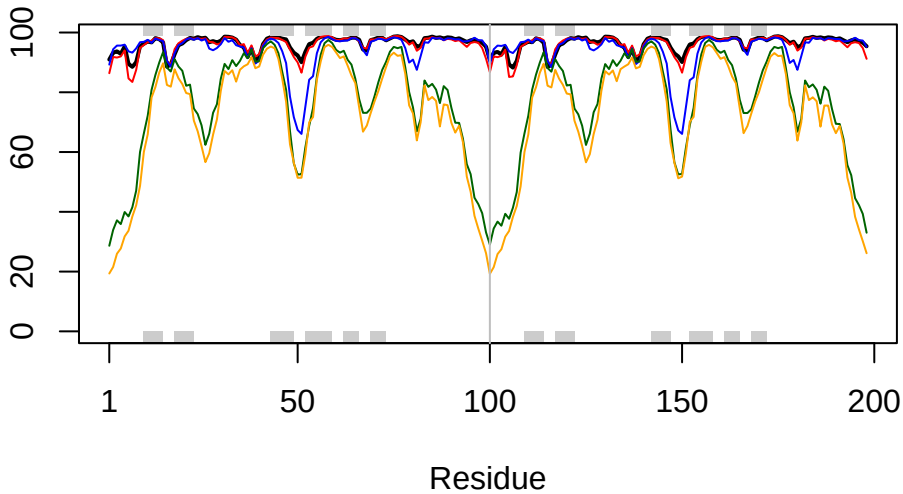
```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```
pdb <- read.pdb("1hsg")
```

```
  Note:  Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9757.214
core size 196 of 198  vol = 8985.811
core size 195 of 198  vol = 3169.051
core size 194 of 198  vol = 1572.467
core size 193 of 198  vol = 1415.614
core size 192 of 198  vol = 1352.407
core size 191 of 198  vol = 1291.262
core size 190 of 198  vol = 1236.417
core size 189 of 198  vol = 1145.775
core size 188 of 198  vol = 1091.912
core size 187 of 198  vol = 1034.344
core size 186 of 198  vol = 982.07
core size 185 of 198  vol = 929.556
core size 184 of 198  vol = 870.277
core size 183 of 198  vol = 832.754
core size 182 of 198  vol = 793.235
core size 181 of 198  vol = 759.982
core size 180 of 198  vol = 725.279
core size 179 of 198  vol = 684.372
core size 178 of 198  vol = 657.54
```

```
core size 177 of 198  vol = 635.321
core size 176 of 198  vol = 615.026
core size 175 of 198  vol = 595.048
core size 174 of 198  vol = 574.548
core size 173 of 198  vol = 558.08
core size 172 of 198  vol = 530.216
core size 171 of 198  vol = 515.328
core size 170 of 198  vol = 500.325
core size 169 of 198  vol = 481.989
core size 168 of 198  vol = 463.138
core size 167 of 198  vol = 447.958
core size 166 of 198  vol = 433.407
core size 165 of 198  vol = 418.55
core size 164 of 198  vol = 407.611
core size 163 of 198  vol = 388.191
core size 162 of 198  vol = 375.704
core size 161 of 198  vol = 364.118
core size 160 of 198  vol = 355.318
core size 159 of 198  vol = 342.769
core size 158 of 198  vol = 330.932
core size 157 of 198  vol = 319.902
core size 156 of 198  vol = 312.329
core size 155 of 198  vol = 301.544
core size 154 of 198  vol = 292.043
core size 153 of 198  vol = 277.975
core size 152 of 198  vol = 264.567
core size 151 of 198  vol = 259.108
core size 150 of 198  vol = 249.631
core size 149 of 198  vol = 238.54
core size 148 of 198  vol = 228.057
core size 147 of 198  vol = 219.476
core size 146 of 198  vol = 209.149
core size 145 of 198  vol = 200.309
core size 144 of 198  vol = 191.697
core size 143 of 198  vol = 181.697
core size 142 of 198  vol = 172.238
core size 141 of 198  vol = 162.501
core size 140 of 198  vol = 152.814
core size 139 of 198  vol = 144.214
core size 138 of 198  vol = 134.532
core size 137 of 198  vol = 129.459
core size 136 of 198  vol = 126.555
core size 135 of 198  vol = 120.409
```

```
core size 134 of 198  vol = 114.556
core size 133 of 198  vol = 108.739
core size 132 of 198  vol = 105.076
core size 131 of 198  vol = 99.07
core size 130 of 198  vol = 94.654
core size 129 of 198  vol = 90.868
core size 128 of 198  vol = 87.652
core size 127 of 198  vol = 83.72
core size 126 of 198  vol = 80.157
core size 125 of 198  vol = 75.296
core size 124 of 198  vol = 69.567
core size 123 of 198  vol = 66.175
core size 122 of 198  vol = 63.445
core size 121 of 198  vol = 60.166
core size 120 of 198  vol = 56.825
core size 119 of 198  vol = 53.932
core size 118 of 198  vol = 51.094
core size 117 of 198  vol = 49.028
core size 116 of 198  vol = 45.741
core size 115 of 198  vol = 41.941
core size 114 of 198  vol = 38.856
core size 113 of 198  vol = 36.504
core size 112 of 198  vol = 33.827
core size 111 of 198  vol = 32.108
core size 110 of 198  vol = 29.123
core size 109 of 198  vol = 27.622
core size 108 of 198  vol = 26.85
core size 107 of 198  vol = 25.83
core size 106 of 198  vol = 24.165
core size 105 of 198  vol = 22.397
core size 104 of 198  vol = 21.963
core size 103 of 198  vol = 22.078
core size 102 of 198  vol = 19.773
core size 101 of 198  vol = 17.479
core size 100 of 198  vol = 15.947
core size 99 of 198  vol = 13.538
core size 98 of 198  vol = 11.818
core size 97 of 198  vol = 10.202
core size 96 of 198  vol = 8.599
core size 95 of 198  vol = 7.179
core size 94 of 198  vol = 5.863
core size 93 of 198  vol = 4.292
core size 92 of 198  vol = 3.546
```

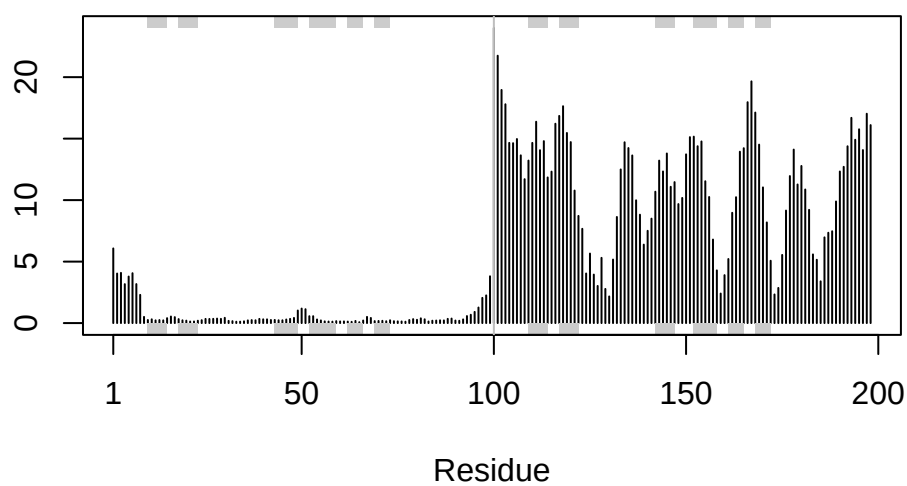
```
 core size 91 of 198  vol = 3.013
 core size 90 of 198  vol = 2.427
 core size 89 of 198  vol = 1.635
 core size 88 of 198  vol = 1.063
 core size 87 of 198  vol = 0.755
 core size 86 of 198  vol = 0.641
 core size 85 of 198  vol = 0.552
 core size 84 of 198  vol = 0.483
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  48     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```
library(jsonlite)

pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt)
```

```
[1] 90.94 93.25 93.69 92.88 95.31 89.50
```

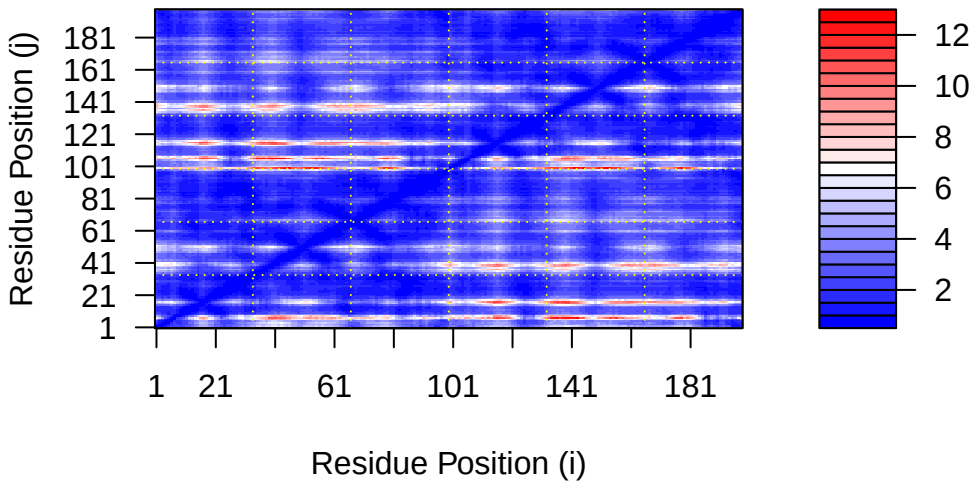
```
pae1$max_pae
```

```
[1] 12.73438
```

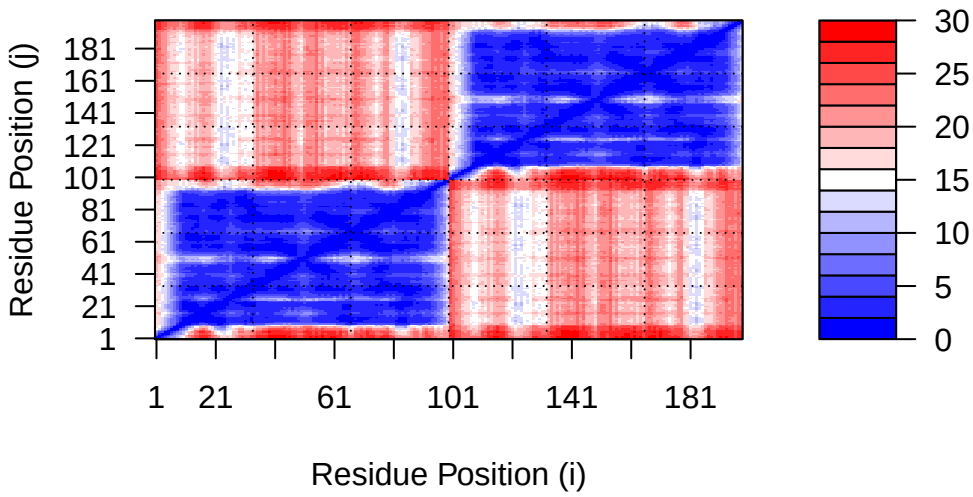
```
pae5$max_pae
```

```
[1] 29.92188
```

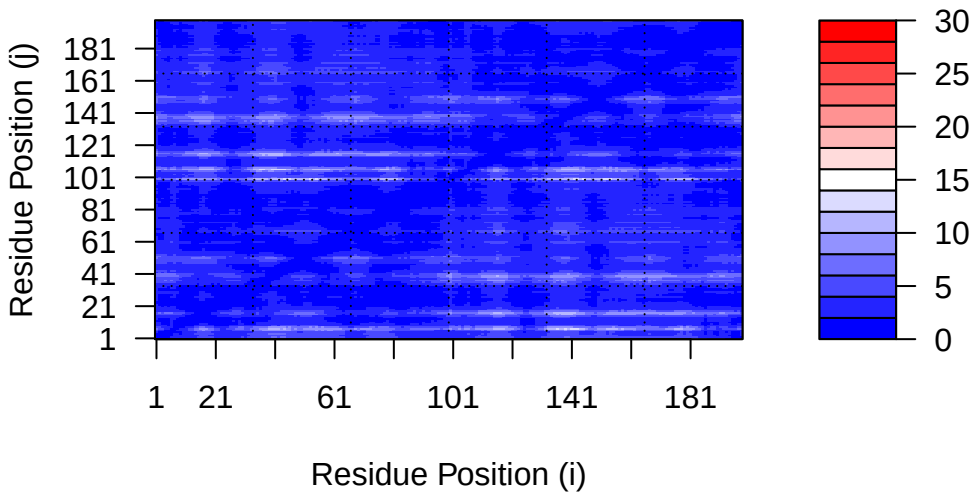
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

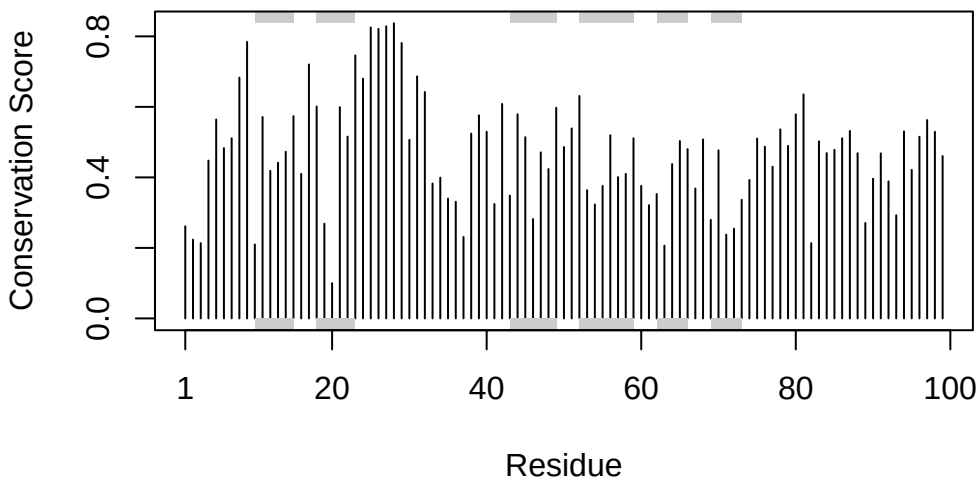
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```